Design Brief – Review 2

0. Header

Project code	FA26SE032
Project name	Sales Collaborator and Affiliate Network Management System (SCANMS)
Team	Nguyễn Thành Thắng (Leader - SE184251), Nguyễn Đình Tuấn (SE182540), Nguyễn Phú Quý (SE180104), Phan Xuân Thịnh (SE184527)
Supervisor	ThS. Tôn Thất Hoàng Minh
Repository link	https://github.com/thang-nt25/SCANMS-project-capstone-fall26 (tag: review-2)
Version	2.0
Date	29/10/2026

1. Scope

1.1 Registration Changes

Summary only; details in Form BM03 (Record of Changes). Write "No changes" if none.

Section	Current	Proposed	Reason
Timeline & Team (Section 1-2)	Duration 09/2026-03/2027; unassigned 5th student row	Duration 07/09/2026-30/12/2026; exactly 4 active members	Align with official university defense calendar; formalize team.
Risks & Security (Section 3.2a, 3.2b)	Generic manual management risks only	Identified return commission overpayment & bot click farming; added Redis sliding-window limit & AI Fraud	Address real e-commerce fraud and attribution spoofing draining budgets.
Multi-Merchant & Logistics (Section 3.2b, 3.2c)	Single-merchant flat catalog; external cart sync; no carrier	Multi-Merchant network, 2-tier Color x Size variant matrix, GHN carrier API integration	Industry standard for multi-store order splitting and delivery tracking.
Roles & Authentication (Section 3.2b, 3.2c)	3 basic roles; allowed unauthenticated guest checkout	5 core roles (added System Manager & Customer); strict authenticated checkout with 0-loss cart	Ensure legal auditability, order provenance, and GHN address validity.
Escrow & Reverse Logistics (Section 3.2b, 3.2c)	Linear payout without quarantine; manual bank export; no returns	Automated 14-day Adaptive Escrow ledger matching customer return warranty; GHN reverse pickup; dual-video dispute center; row-level locks; 10% PIT tax withholding	Prevent merchant deficit and creator wallet overdraft on return disputes.
Collaboration, Live & AI (Section 3.2b, 3.2c)	Static 1-way media download; no messaging; no sample flow; no AI	Real-time Socket.IO deal chat; 14-day sample SLA; live deal pinning; AI Partner Matching & Anti-Fraud	Expand social commerce capabilities and elevate technical capstone rigor.

1.2 Out of Scope

Item	Reason
Proprietary Delivery Fleet & Physical Couriers	Delegated to Giao Hàng Nhanh (GHN) Sandbox API with automatic shipping fee calculation and parcel tracking.
Direct Fiat Commercial Banking Gateway	Delegated to PayOS / VietQR payment gateway sandbox; interbank license is out of university capstone scope.
On-premise Video Transcoding CDN Infrastructure	Delegated to WebRTC peer streaming and simulated HLS chunk distribution for livestream shopping rooms.
Enterprise ERP / WMS Physical Warehouse System	Platform manages digital inventory, variant matrices, and logistics orders without proprietary robotics hardware.

2. System at a Glance

Maximum 150 words: the problem, the users, the core value. Link to the architecture diagram in the SDD.

3. Key Design Decisions

Exactly 5 decisions. Each must have one owner.

ID	Decision	Options considered	Chosen option and rationale	Owner	Related FR / diagram
D1	Additive Multi-Rule Commission Engine	Uncapped linear sum vs flat single rate	Formula Rate = min(25%, Base + Tier + max(Deal, Campaign) + Live) balances creator incentives with merchant margin safety.	Nguyễn Đình Tuấn	FR-09 / SDD Section 7
D2	Strict Authenticated Checkout with Zero-Loss Cart	Anonymous guest checkout vs forced pre-login	Enforced authenticated purchase with local storage cart preservation, guaranteeing KYC auditability and GHN address validity.	Nguyễn Thành Thắng	FR-16 / SDD Section 6
D3	14-Day Adaptive Escrow Commission Holding Ledger	Instant payout upon delivery vs fixed 7-day vs 14-day adaptive ledger	14-day post-delivery quarantine ledger matching customer 14-day return SLA with automated return clawback, preventing creator wallet overdraft during return window.	Nguyễn Phú Quý	FR-21 / SDD Section 5
D4	3-Tier Multi-Store Voucher Isolation Hierarchy	Single cart-wide voucher vs merchant-only coupon	Platform global + Store (couponsByStore) + Product voucher hierarchy with atomic order splitting and pre-checkout margin validation.	Nguyễn Đình Tuấn	FR-12 / SDD Section 4
D5	Dual-Channel Real-time Live & Messaging Gateway	HTTP polling vs external commercial live SaaS	Decoupled Socket.IO WebSocket gateways for live deal pinning and chat, with local memory fallback under Redis degradation.	Phan Xuân Thịnh	FR-26 / SDD Section 3

4. Top Risks

ID	Risk	Impact	How we verify or mitigate	Owner
R1	Attribution Spoofing & Click Spamming draining merchant affiliate budgets.	High: Financial distortion and collaborator attribution disputes.	Sliding-window Redis rate limiter (100 req/min/IP), device fingerprinting, and automated AI fraud anomaly detection (FR-31).	Nguyễn Thành Thắng
R2	Double-Discount Margin Erosion from stacked vouchers and affiliate rates.	High: Merchant losses on subsidized high-commission campaign sales.	Pre-checkout margin protection guard enforcing total discount <= allowed profit margin, plus 25% global commission cap.	Nguyễn Đình Tuấn
R3	Collaborator Sample Exploitation violating 14-day review content delivery SLA.	Medium: Merchant inventory loss and unfulfilled marketing campaigns.	Automated 14-day SLA deadline tracking, reminder notifications, sample quota throttling, and System Manager dispute blacklisting.	Phan Xuân Thịnh

5. Known Gaps

Mismatches between the registration form, SRS, SDD and code. Include every High finding marked Kept or Planned in Form BM02.

ID	Gap	Location	Plan	Owner	Due
G1	Automated Fiat Interbank Payout Disbursal vs Sandbox Simulation	SDD Section 6 / payouts.service.ts	Implemented PayOS sandbox payment adapter; production bank host-to-host switch ready pending enterprise corporate KYC.	Nguyễn Phú Quý	Review 3
G2	In-browser WebRTC Peer Streaming vs Commercial CDN Media Cluster	SDD Section 5 / live-sessions.service.ts	Socket.IO signaling and HLS simulation tested up to 50 concurrent viewers; cloud CDN cluster deferred to commercial deployment.	Phan Xuân Thịnh	Council 1.1
G3	Distributed Redis SPOF Graceful Degradation	SDD Section 3 / cache.service.ts	Implemented in-memory LRU Map fallback with automatic reconnection retry loop and operational degraded alerts (Fixed & verified in CI).	Nguyễn Đình Tuấn	Done

6. Proposed Mandatory Features

Within the scope in section 1. Acceptance criteria must be testable with a clear pass/fail result.

ID	Feature	Actor	Acceptance criteria (testable)	Test	Status
F-01	User Authentication & RBAC Guard	Customer, Collaborator, Shop Manager, System Manager, System Administrator	Argon2id password hashing, JWT Access/Refresh tokens, strict RBAC guard for 5 roles.	TC-F01-01..03	**Done**
F-02	Identity Verification & KYC Multi-Tier Approval	Collaborator, Shop Manager, System Manager	CCCD / Business license upload, partner upgrade workflow (/customer/upgrade), multi-status verification.	TC-F02-01..03	**Done**
F-03	Collaborator Social Portfolio Linking	Collaborator	Link TikTok, Facebook, YouTube channels with follower counts and content category tags.	TC-F03-01..03	**Done**
F-04	Digital Media Asset Hub	Shop Manager, Collaborator	Centralized promotional asset library with tags, resolution filtering, and one-click download.	TC-F04-01..03	**Done**
F-05	Real-time Security Audit Logging	System Administrator, System Manager	Immutable recording of sensitive financial, KYC, payout, and dispute actions with actor ID & IP.	TC-F05-01..03	**Done**
F-06	Additive Multi-Rule Commission Engine	Shop Manager, Collaborator, System Manager	Calculates min(25%, Base(10%) + Tier(1-3%) + max(Deal, Campaign) + Live(2-3%)) + MonthlyBonus.	TC-F06-01..03	**Done**
F-07	Smart Shortened Link & QR Generator	Collaborator	Generates /r/:code links and SVG dynamic QR codes with Redis rate limit protection.	TC-F07-01..03	**Done**
F-08	Last-Click Cookie Attribution & Tracking	Customer, Collaborator	30-day cookie window, first/last touch attribution, referrer preservation across sessions.	TC-F08-01..03	**Done**
F-09	3-Tier Multi-Store Voucher Stacking	Customer, Shop Manager	Stack Global + Store (couponsByStore) + Product vouchers with pre-checkout margin check.	TC-F09-01..03	**Done**
F-10	Multi-Store Order Splitting & In-place Quick Login	Customer	Atomic DB order splitting per store, zero-loss login preservation without cart redirect, rollback on failure.	TC-F10-01..03	**Done**
F-11	6-Block Merchant Product Variant Matrix	Shop Manager, System Manager	Shopee-standard 6-block layout: basic info, media, 2-tier variant matrix (Color x Size), logistics, commission, legal.	TC-F11-01..03	**Done**
F-12	Merchant Order Lifecycle & GHN Fulfillment	Shop Manager, Customer	GHN Sandbox parcel creation, dynamic weight/dimension fee calculation, live webhook tracking.	TC-F12-01..03	**Done**
F-13	14-Day Adaptive Escrow Commission Holding Ledger	Shop Manager, Collaborator, System Manager	Holds commission in escrow post-delivery for 14 days + Vietnam holidays; automated release and return clawback.	TC-F13-01..03	**Done**
F-14	Return, Refund & Dual-Video Dispute Mediation Center	Customer, Shop Manager, System Manager	14-day return flow, shop warehouse setup, GHN reverse pickup, 48h shop inspection SLA, dual-video arbitration.	TC-F14-01..03	**Done**
F-15	Collaborator Sample Request & 14-Day SLA	Collaborator, Shop Manager, System Manager	Sample product request, quota throttling, GHN delivery tracking, 14-day review content proof SLA.	TC-F15-01..03	**Done**
F-16	Real-time Chat & Exclusive Deal Negotiation	Collaborator, Shop Manager	Socket.IO real-time chat, rich proposal card, bi-directional exclusive commission negotiation.	TC-F16-01..03	**Done**
F-17	Interactive Livestream Shopping Room	Collaborator, Customer	WebRTC/HLS room, dynamic live deal pinning, real-time live voucher drop, conversion metrics.	TC-F17-01..03	**Done**
F-18	Realtime Analytics & Leaderboard with AI Fraud	Collaborator, Shop Manager, System Manager, System Administrator	Real-time GMV/commission dashboards, gamified tier ranking, AI traffic fraud sentinel.	TC-F18-01..03	**Done**
F-19	Concurrency-Safe Wallet Withdrawal & Payout Management	Collaborator, Shop Manager, System Manager	Automated withdrawal request with DB row-level locking, 10% PIT tax withholding, VietQR batch export, and bank bill verification.	TC-F19-01..03	**Done**

7. Non-Functional Requirements

Maximum 6 rows. Each NFR needs a metric, a threshold and a measurement method.

ID	Quality attribute	Metric	Threshold	Measurement method
NFR-01	Performance & Latency	API Response Time (p95)	<= 200 ms for catalog browsing and tracking redirection	Autocannon load test (1,000 req/s) & NestJS duration interceptor
NFR-02	Security & Password Hashing	Argon2id Hash & JWT Lifespan	Argon2id (>= 64MB memory, 3 iterations); JWT access token 15 min	Automated auth test suites (auth-otp.spec.ts) & OWASP ZAP scan

ID	Quality attribute	Metric	Threshold	Measurement method
NFR-03	Availability & Transaction Safety	Uptime & Atomic Rollback Rate	99.5% service uptime; 100% atomic rollback on stock exhaustion	Jest E2E order rollback simulation & /health endpoint monitoring
NFR-04	Frontend Web Vitals	Bundle Size & First Contentful Paint	Initial bundle <= 500 KB (brotli); FCP <= 1.2s; CLS < 0.1	Vite build bundle visualizer & Google Lighthouse CI audit
NFR-05	Concurrency & Rate Limiting	Sliding Window Limit & Failover	100 req/min/IP with zero crash on Redis outage (RAM fallback)	Jest unit safety tests (jest-unit-safety.spec.ts) & Apache Bench
NFR-06	Financial Data Auditability	Audit Log Completeness	100% of financial, commission, and dispute movements logged	Prisma audit interceptor verification test (audit.interceptor.spec.ts)

8. Ownership Map

One row per member. Each member owns 2–3 parts and at least one main workflow (Appendix A1).

Member	Role	Owned parts (2–3)	Main workflow owned	Key decisions / PRs
Nguyễn Thành Thắng (Leader)	Fullstack Architect & Security Lead	Auth & RBAC (F-01), KYC Engine (F-02), Social Portfolios (F-03), Media Hub (F-04), Audit Sentinel (F-05)	WF-01: User Registration, 2FA Authentication & Multi-tier KYC Approval	PR #1 (Auth & RBAC), PR #5 (KYC Engine), Decision D2
Phan Xuân Thịnh	Fullstack Developer & Realtime Lead	Commission Rates (F-06), Sample Request SLA (F-15), Real-time Deal Chat (F-16), Livestream Shopping Room (F-17)	WF-02: Exclusive Deal Negotiation, In-App Chat & Live Commerce Deal Pinning	PR #16 (Live Streaming Room), PR #27 (Chat Socket.IO), Decision D5
Nguyễn Đình Tuấn	Backend Lead & Financial Engine Specialist	Smart Link & Dynamic QR (F-07), Last-Click Cookie (F-08), Voucher Stacking (F-09), Order Splitting & Checkout (F-10), Analytics & AI Fraud (F-18), Concurrency Payouts (F-19)	WF-03: Multi-Channel Attribution, Quick Login Modal & VietQR Checkout; WF-05: Multi-Role Wallet Withdrawal & Payouts (Co-owner)	PR #9 (Commission Engine), PR #12 (Voucher Stacking), Decisions D1 & D4
Nguyễn Phú Quý	Fullstack Developer & Logistics Lead	Product Variant Matrix (F-11), Order Lifecycle & GHN (F-12), 14-Day Adaptive Escrow (F-13), Returns & Dispute Mediation (F-14)	WF-04: GHN Logistics Fulfillment, 14-Day Adaptive Escrow & Dual-Video Return Arbitration	PR #11 (Product Matrix), PR #14 (Return State Machine), Decision D3

Appendix A – Defense 1.1 Targets

A1. Main Workflows

Minimum number = number of team members. A valid workflow involves at least 2 actors or at least 3 mandatory features, and changes the state of at least one main entity.

ID	Workflow	Actors	Trigger	Business outcome	Features	Entity states	Owner
WF-01	User Registration, 2FA Authentication & Multi-tier KYC Approval	Collaborator, Shop Manager, System Manager	Collaborator or Shop Manager registers account, verifies OTP, and submits identity dossier	Verified user account with role-based portal access, KYC verified badge, full affiliate capabilities unlocked	F-01, F-02, F-03, F-04, F-05	User: PENDING -> ACTIVE; KYC: PENDING_REVIEW -> VERIFIED; Tier: SILVER	Nguyễn Thành Thắng
WF-02	Exclusive Deal Negotiation, In-App Chat & Live Commerce Deal Pinning	Collaborator, Shop Manager, Customer	Shop Manager publishes 10% base product; Collaborator proposes 13% VIP rate via Chat OR Shop Manager invites Collaborator; Collaborator starts live session	VIP Deal agreed, VIP Voucher Code & Smart Link issued, live session with deal pinning (+3% live bonus, e.g. 16%)	F-06, F-15, F-16, F-17	Deal: PROPOSED -> AGREED; LiveSession: SCHEDULED -> LIVE -> ENDED; Pin: PINNED -> UNPINNED	Phan Xuân Thịnh
WF-03	Multi-Channel Attribution, Quick Login Modal & VietQR Checkout	Customer, Collaborator, Shop Manager, Payment Gateway	Customer clicks Collaborator Smart Link / QR / Live Deal, adds multi-store items, and clicks Checkout	30-day marketing cookie recorded, 100% cart preserved during in-place login modal, instant VietQR payment verification, order PAID	F-06, F-07, F-08, F-09, F-10	Link: ACTIVE; Cookie: SET (30d); Order: PENDING_PAYMENT -> PROCESSING (PAID); Cart: PRESERVED	Nguyễn Đình Tuấn
WF-04	GHN Logistics Fulfillment, 14-Day Adaptive Escrow & Dual-Video Return Arbitration	Shop Manager, Customer, GHN Logistics, System Manager	Shop Manager confirms order, creates GHN shipment parcel, delivers to Customer; Customer requests return within 14 days	GHN waybill generated, delivery completed, 14-day adaptive escrow quarantine active; if return disputed: dual-video arbitration, 100% Customer wallet refund	F-11, F-12, F-13, F-14	Order: PROCESSING -> SHIPPED -> DELIVERED -> COMPLETED; Escrow: LOCKED -> RELEASED/CLAWED_BACK	Nguyễn Phú Quý
WF-05	Multi-Role Wallet Withdrawal, 10% PIT Tax Withholding & Concurrency Payout Settlement	Collaborator, Shop Manager, System Manager, Banking Gateway	Collaborator or Shop Manager submits withdrawal request, or month-end bonus scan calculates GMV milestone	10% statutory PIT tax withheld, available balance locked via database row-level locking, VietQR batch exported, bank bill verified, and funds disbursed via Napas 24/7	F-05, F-06, F-18, F-19	Wallet: LOCKED -> DEBITED; Withdrawal: PENDING -> PROCESSING -> APPROVED -> DISBURSED; Tax: WITHHELD	Nguyễn Đình Tuấn & Nguyễn Thành Thắng

A2. Demo Scenarios

Per workflow: at least 1 main scenario and 1 exception scenario. Run on staging with seeded data.

ID	Workflow	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-01a	WF-01	**Main**	TC_KYC_01..04: Collaborator or Shop Manager registers with phone/email, verifies	1. Register account with Email & Phone. 2. Submit 6-digit OTP. 3. Upload CCCD &	Account status ACTIVE; KYC badge marked VERIFIED; role portal access unlocked (TC_KYC_01..04).	Nguyễn Thành Thắng

ID	Workflow	Type	Preconditions and data	Main steps	Expected result	Presenter
			OTP, submits 12-digit ID with 2 photos.	Bank details. 4. System Manager audits & approves.		
DS-01b	WF-01	**Exception**	TC_KYC_01..04: Duplicate email / weak password / >5 wrong OTP locks 15m OR blurred ID photo.	1. Submit duplicate email. 2. Input wrong OTP 5 times. 3. System Manager clicks Reject with mandatory feedback note.	Account locked 15 mins against brute-force; KYC transitions to REJECTED; resubmission unlocked (TC_KYC_01..04).	Nguyễn Thành Thắng
DS-02a	WF-02	**Main**	TC_DEAL_01..04: Shop Manager proposes 13% VIP rate via Chat; Collaborator clicks Accept; Collaborator goes LIVE and pins deal.	1. Shop Manager selects product, proposes 13% rate in Chat. 2. Collaborator clicks Accept Deal. 3. Collaborator goes LIVE. 4. Pins deal (+3% live bonus).	VIP voucher generated; live stream active with +3% live bonus (16% total rate); summary report shown on stream end (TC_DEAL_01..04).	Phan Xuân Thịnh
DS-02b	WF-02	**Exception**	TC_DEAL_01..04: Proposal rate > 25% ceiling blocked; Collaborator rejects deal (retains base 10%); live item out of stock.	1. Input rate > 25%. 2. Collaborator clicks Reject. 3. Product stock depleted to 0 during live broadcast.	Submission blocked by 25% ceiling cap; deal canceled retaining base 10%; purchase button automatically hidden in live tray (TC_DEAL_01..04).	Phan Xuân Thịnh
DS-03a	WF-03	**Main**	TC_ATTR_01..04: Collaborator generates /r/:slug smart link; Customer clicks (cookie set 30 days); adds multi-store items; checks out.	1. Collaborator generates smart link & QR. 2. Customer clicks link. 3. Adds items from Shop A & B. 4. Quick login in-place. 5. Scans VietQR.	30-day marketing cookie recorded; 100% cart preserved during login modal; PayOS webhook marks order PAID (TC_ATTR_01..04).	Nguyễn Đình Tuấn
DS-03b	WF-03	**Exception**	TC_ATTR_01..04: Unverified KYC blocks link creation; Customer visits deleted item (404); VietQR underpaid.	1. Unverified Collaborator attempts link creation. 2. Customer visits deleted SKU. 3. Customer underpays VietQR invoice.	Link button disabled with KYC prompt; 404 Discontinued page displayed; underpaid order remains in Pending Payment (TC_ATTR_01..04).	Nguyễn Đình Tuấn
DS-04a	WF-04	**Main**	TC_FUL_01..04: Shop Manager creates GHN parcel with valid weight/dimensions; tracking code issued; parcel delivered.	1. Shop Manager packages order. 2. Confirms weight & dimensions. 3. Calls GHN Sandbox API. 4. Delivery completed.	GHN tracking code generated (e.g. GHN-TEST-12345); status In Transit -> Delivered; 14-day adaptive escrow quarantine active (TC_FUL_01..04).	Nguyễn Phú Quý
DS-04b	WF-04	**Exception**	TC_FUL_01..04: Customer submits 14-day return claim with unboxing video; Shop Manager inspection SLA 48h; System Manager arbitrates.	1. Customer requests return with video. 2. Shop Manager rejects dispute. 3. System Manager reviews unboxing vs packaging videos.	Escrow frozen; System Manager approves Customer claim: 100% wallet refund, commission clawed back (TC_FUL_01..04).	Nguyễn Phú Quý
DS-05a	WF-05	**Main**	TC_PAY_01..04: Collaborator or Shop Manager requests payout; 10% PIT tax withheld; balance locked via DB row-level lock; System Manager audits recipient bank account, uploads bank transfer bill, and dispatches via Napas 24/7.	1. Collaborator or Shop Manager submits withdrawal request. 2. 10% PIT tax withheld and available balance locked. 3. System Manager reviews bank account details and attaches bank transfer receipt. 4. Dispatches payout via Napas 24/7.	Net amount disbursed to bank; 10% PIT tax withheld; transaction receipt attached; locked balance permanently debited (TC_PAY_01..04).	Nguyễn Đình Tuấn & Nguyễn Thành Thắng
DS-05b	WF-05	**Exception**	TC_PAY_01..04: Insufficient balance (< 100k or > balance); unauthorized access; invalid bank account; bank gateway failure.	1. Request amount > available balance. 2. Unauthorized role attempts approval. 3. Bank gateway returns transfer failure.	Submission blocked; 403 Forbidden on unauthorized audit; 100% requested funds safely restored to wallet with error logged (TC_PAY_01..04).	Nguyễn Đình Tuấn & Nguyễn Thành Thắng

A3. Functional Test Case Target

Source	Count	Minimum per item	Subtotal
Mandatory features (from Section 6)	19	3	57
Main workflows (end-to-end, from A1)	5	1	5
State transitions of main entities (Order, Sample, KYC, Payout)	13	1	13
Agreed NFRs & Security suites (from Section 7)	6	1	6

Source	Count	Minimum per item	Subtotal
Committed total (minimum)	**43**	**—**	**81**

Automated in CI: at least 50% of the committed total.